John Grace

Education

MSc. Physics
ETH Zurich, Switzerland
2026 – 2028
Relevant Coursework: Quantum Field Theory I, Quantum Information Processing I&II, Symmetries, Foundations of General Relativity

BSc. Honours Physics
McGill University, Canada
2021 – 2025
GPA: 3.95/4.00
Recognition: Distinction, First Class Honours, Math and Physics "Class of '65" Award
Relevant Coursework: Honours Quantum Mechanics I&II, Quantum Condensed Matter, Applied Machine Learning, General Relativity
Thesis: Double Homodyne Detection of Lossy Squeezed Vacuum and Coherent Light for Quantum Interferometry (Sup. by Prof. Bill Coish and Prof. Kai Wang)

Experience

Research Assistant
National Research Council of Canada
Summer 2026
- Implementing a quantum non-demolition experiment in the weak and strong measurement regimes to create a robust measurement protocol for practical quantum networks.
- Supervised by Prof. Jeff Lundeen.

Research Assistant
University of Naples Federico II
Autumn 2025
- Lead an experimental effort to push classical structured media to a quantum regime, incorporating the use of a SPAD-array camera to do space-resolved coincidence measurements of single photons.
- Applied machine learning methods for the inverse design of quantum metasurfaces. Designed high-dimensional optical gates on the transverse momentum of light.
- Supervised by Dr. Francesco DiColandrea and Prof. Fillipo Cardano.

Undergraduate Researcher
McGill University
Autumn 2023 - Summer 2025
- My honours thesis studied the upper bound of the precision in phase estimation using quantum states of light with practical considerations (photon loss, phase variance).
- Undertook a short research project in building an synthetic frequency dimension using optical ring resonators for the simulation of lattice physics.
- Supervised by Prof. Bill Coish and Prof. Kai Wang.

Undergraduate Researcher
University of Ottawa
Summers 2023 & 2024
- Studied how optical metasurfaces can be used in a variety of applications, including the tomography of photonic qubits, engineering open-system quantum dynamics, and the simulation of topological quantum systems.
- Supervised by Dr. Francesco DiColandrea and Prof. Ebrahim Karimi.

Academic Publications

- **Engineering qubit dynamics in open systems with photonic synthetic lattices** | Phys. Rev. Res. **7**, 023236 (2025)
- **Single-acquisition tomography of photonic qubits with structured media** | arXiv:2607.03052 (2026)

Workshops & Conferences

- **Photonics North** | Montreal, Canada | *2023*
- **Light-Matter Interaction Mini-Workshop** | University of Ottawa | *2023*
- **RQMP Grande Conference** | Sherbrooke, Canada | *2023*
- **Max Planck Institute for the Study of Light - Annual Meeting** | University of Ottawa | *2024*
- **INTRIQ Meeting** | Sherbrooke, Canada | *2024*
- **Perimeter Scholars International 'PSI-Start' Satellite Program** | Bishop's University | *2025*
  I was one of ten students invited to partake in a two-week summer school in modern theoretical physics. Topics included: Path integrals, quantum information & thermodynamics, symmetries, and numerical methods in condensed matter physics.
- **Max Planck Institute for the Study of Light - Annual Meeting** | University of Ottawa| *2026*
  Presented a poster on 'Single-acquisition tomography of photonic qubits with structured media'.

Language Skills

- **English** | Native
- **French** | Proficient
- **German** | Beginner